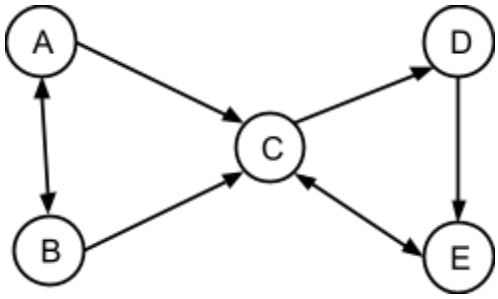
American Computer Science League

2025-2026 ● Contest 4: Short Problems ● Junior Division

1. Graph Theory

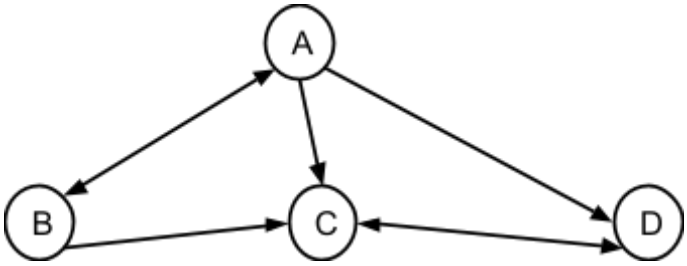
How many paths of length 2 are there in the following graph?



A. 6
B. 8
C. 9
D. 11
E. 13

2. Graph Theory

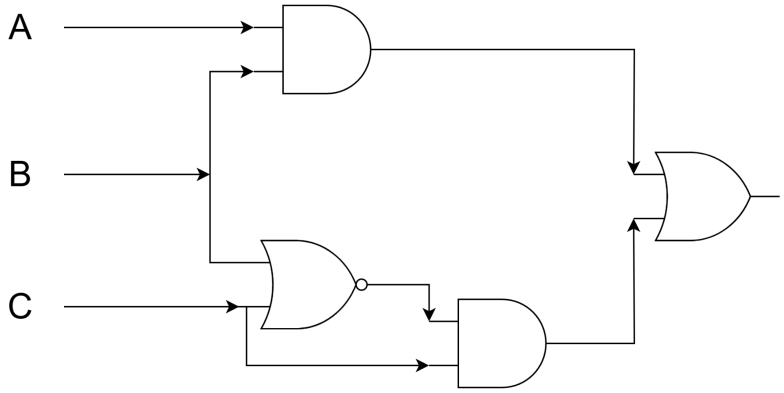
How many cycles are there in the directed graph below?



A. 0
B. 1
C. 2
D. 3
E. 4

3. Digital Electronics

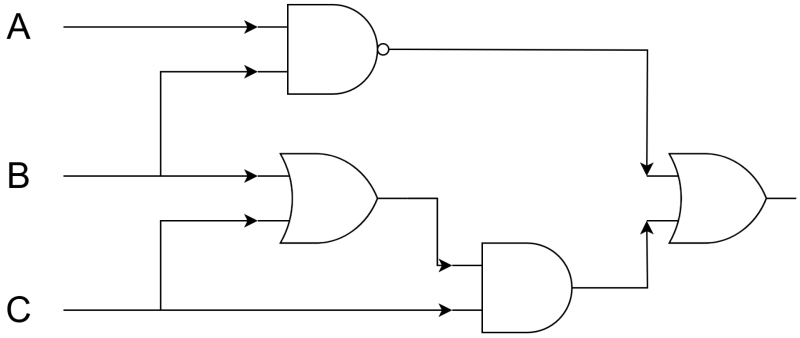
Simplify the Boolean algebra expression represented by the following digital circuit:



A. $\overline{A}B$
B. $A\overline{B}$
C. $A + B$
D. AB
E. $AB + \overline{B}C$

4. Digital Electronics

How many ordered triples make the following digital circuit TRUE?



A. 1
B. 3
C. 5
D. 7
E. 8

5. What Does This Program Do? (Strings)

What will be the output for the following program if the function val(*s*) converts the string *s* to its numerical value and str(*a*) converts the numeric value *a* to a string?

```
s = "742"
r = ""
for x = 0 to 2
     r = s[x] + r
next x
a = str(val(s) - val(r))
b = ""
for x = len(a)-1 to 0 step -1
     b = b + a[x]
next x
output val(b) + val(a)
```

A. 99
B. 189
C. 989
D. 1089
E. 495954

6. What Does This Program Do? (Strings)

What will be the output for the following program given that all uppercase letters come before all lowercase letters alphabetically?

```
s = "APCompSciAinJava"
ans = ""
for x = 0 to len(s) - 2 step 2
     if s[x] < s[x+1] then
          ans = ans + s[x] + s[x+1]
     else
          ans = ans + s[x+1] + s[x]
     end if
next x
output ans
```

A. APCompSciAinJava
B. APCompScAiinJaav
C. APCOMPSCAiinJava
D. PACOMPSCAiinJava
E. PAoCpmcSiAniaJva